

Venkata Naga Sai Bhavan Koppolu

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SUMMARY

Master's student in Robotics & Autonomous Systems, Artificial Intelligence at Arizona State University with hands-on experience spanning AI research, applied machine learning, robotics, and industry software and system development. Proven ability to design, implement, and evaluate end-to-end AI and robotic systems across NLP, computer vision, generative AI, and autonomous systems. Experienced in taking projects from research and experimentation through to client-facing deployment, with a strong foundation in both academic rigor and practical engineering.

EDUCATION

Arizona State University

M.S. in Robotics and Autonomous Systems, Artificial Intelligence

Tempe, AZ

Jan. 2025 – Dec. 2026

Queen Mary University of London

B.S. in Computer Science

London, UK

Sept. 2021 – Aug. 2024

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, R, SQL, MATLAB, Bash

AI, ML & Generative AI: PyTorch, TensorFlow, Hugging Face, JAX, scikit-learn, Supervised Learning, Fine-Tuning, RAG, Agentic AI, Quantization, LLM/vLLM, LangChain, Diffusion Models, RLHF, Instruction Tuning

NLP & Computer Vision: Multilingual NLP, Low-Resource NLP, Transformers, Text Classification, Object Detection, Image Segmentation, OpenCV, Sentiment Analysis, Named Entity Recognition

Robotics & Autonomous Systems: ROS/ROS2, Motion Planning, Inverse Kinematics, Perception-Planning-Action Pipelines, Reinforcement Learning, Sensor Fusion, PID Control, Isaac Sim, Gazebo, AWS DeepRacer

Cloud, HPC & MLOps: AWS, GCP, Azure, Docker, Kubernetes, SLURM, HPC/Supercomputing, Distributed Training, MLflow, Model Parallelism, Weights Biases, CI/CD

Web, Data & Tools: React, Node.js, Spring Boot, APIs Calling, NumPy, Pandas, Matplotlib, MySQL, Git, Linux

EXPERIENCE

AI/ML Systems Engineering Intern

DeC. 2025 – Present

Innominds

Remote

- Engaged directly with Fortune 100 clients to scope, develop, and present AI and automation solutions, translating complex technical architectures into actionable business use cases and POC deliverables.
- Built an Agentic BI POC for Novartis combining a NetworkX/Pyvis knowledge graph, an autonomous pipeline monitoring engine, and an NLP query interface (Llama 3.2, LangChain) over synthetic molecular R&D data; delivered live Colab demo to enterprise stakeholders.
- Authored detailed AI annotation pipeline specifications for a Qualcomm autonomous driving dataset, defining transformer-based 3D LiDAR pre-annotation (Point Transformer v2, 3DETR), multi-sensor calibration validation (LiDARcamera extrinsics, reprojection residuals, ego-pose drift), confidence-based human-in-the-loop routing, and versioned ground truth workflows across nuScenes and aiMotive datasets.

Software Engineering Intern

June 2023 – Sept. 2023

Techwish

Hyderabad, India

- Developed backend microservices using Java, Spring Boot, Hibernate, and MySQL for live projects serving Fortune 100 clients.
- Built responsive HTML/CSS frontend components and integrated REST APIs to support user-facing functionality
- Developed JUnit test cases, used Git/GitHub version control, and contributed across the Software Development Life Cycle (SDLC).

Software Engineering Intern

June 2022 – Sept. 2022

Innominds

Hyderabad, India

- Developed full-stack features across Java, .NET, and ASP.NET codebases on live projects under senior Technology Leaders.
- Built responsive HTML/CSS UI components to improve frontend functionality and backend service integration.
- Contributed to Agile sprints through requirements gathering, client interactions, and technical documentation.

Lightweight Martian Terrain Classification for Rover Safety

Jan. 2026 – Apr. 2026

Manuscript in preparation for submission to ICRA, IROS, i-SAIRAS

- Developed lightweight perception systems (Tiny CNN, SNN) optimized for edge deployment, benchmarking models across the 16,064-image AI4MARS dataset to meet strict rover-class hardware constraints.
- Evaluated heterogeneous compute performance using CodeCarbon, verifying a 0.025 ms/image inference latency and a minimal 1.07M parameter footprint.
- Mapped perception workloads onto neuromorphic architectures, validating an 83.3% test accuracy and demonstrating a 52% theoretical energy advantage for SNNs.

Memorization in Diffusion Models

Jan. 2026 – Apr. 2026

- Conducted an experimental study on memorization behavior in diffusion models using a simplified DDPM U-Net architecture trained on CIFAR-10.
- Designed and evaluated experiments analyzing the effects of conditioning type, dataset size, training duration, and label specificity on memorization behavior.
- Implemented diffusion training, sampling, and memorization evaluation pipelines in PyTorch, utilizing A100 GPUs across Google Colab and the ASU Sol Supercomputer, reproducing memorization rates of up to 99.8% under controlled conditions.

Identifying LLM Performance Gaps Across Languages

Aug. 2025 – Dec. 2025

- Evaluated OLMo-2-7B-Instruct across 100+ languages and 10+ benchmarks using vLLM and HPC infrastructure, establishing multilingual performance baselines.
- Built a multilingual translation pipeline using NLLB-200 and Hunyuan-MT-7B, validating outputs with BLEU, METEOR, COMET, and LaBSE metrics.
- Quantified token exposure vs. accuracy (Spearman's ρ of 0.94), identifying data imbalance as the main driver of multilingual performance gaps.

Agentic AI Robotic Puzzle-Solver

Aug. 2025 – Dec. 2025

- Integrated perception with planning and control layers by developing an autonomous Vision-Language-Action (VLA) framework using Google Gemini 2.5 Flash for a physical robotic manipulator.
- Designed semantic mapping and environment representations, utilizing 3x3 Affine Transformations to map 2D OpenCV bounding boxes into depth-aware 3D spatial coordinates.
- Implemented LLM function calling to abstract complex spatial reasoning into Python motion controls, mitigating real-world constraints like visual hallucinations to ensure deterministic kinematic execution.

AWS DeepRacer

Aug. 2025 – Dec. 2025

1st Place, AWS DeepRacer Contest

- Engineered a geometry-aware reward function with curvature-sensitive speed shaping, applying policy gradient principles to replace a naive centering reward that caused instability and suboptimal convergence on tight turns.
- Redesigned the discrete action space to decouple speed and steering, reducing state-action coupling and improving exploration efficiency across the policy's decision boundary.
- Applied curriculum learning and hyperparameter tuning across progressively harder track configurations, achieving significant lap time reduction and first-place finish in the final contest evaluation.

Using NLP to Detect Fake Reviews in Telugu

Sept. 2023 – Apr. 2024

*4th Place, Top 10 Most Innovative Undergraduate Projects 2024**Manuscript in preparation for submission to ACL, EMNLP*

- Engineered a novel low-resource NLP dataset by translating the 1,600-review OpSpam corpus into Telugu, establishing a critical foundation for cross-lingual deception detection.
- Designed comprehensive transfer learning and zero-shot experiments using mBERT, systematically tuning training hyperparameters (batch size, learning rates) via grid search.
- Demonstrated that a 50/50 mixed-language training composition achieves 75.9% accuracy – nearly matching the 79.1% monolingual baseline – providing actionable efficiencies for training LLMs in linguistic domains.

FDM Wellbeing App

Jan. 2023 – Apr. 2023

- Engineered CI/CD test automation pipelines for an enterprise-grade web application using ReactJS and Node.js, systematically validating 6+ complex software workflows.
- Architected and verified backend database infrastructure by testing and linking 25+ API endpoints, ensuring continuous delivery across the development lifecycle.
- Improved end-to-end system efficiency by deploying automated acceptance testing across 15+ core use cases, eliminating flaky builds and validating predictable performance.